

Bench Run — Teach and Playback

Prove a step is recorded when you let go, build a park pose, and play it back

Audience: the person at the bench, one hand free for the rocker | **Arm:** Emre Kalem 6-axis, 4× MG996R + 3× MG90S

Controller: Arduino Uno R3 on the Pi `mira-arm`, `/dev/ttyACM0` | **Firmware:** `FACTORYLM-ARM PROTO=1.0 FW=1.1.1 BUILD=20260808` [`factorylm_arm_controller.ino` line 566]

Console: `1.3.0` — tabbed pendant layout, auto-step recording, OUT OF RANGE verdict [`arm-console.html` `CONSOLE_VERSION`; commits 2630d85, 1b4fc41, 1565b44]

Start it with: the **Arm Console** icon on the desktop | **Long form:** `Documentation/NEXT-BENCH-RUN.md`, `Documentation/RECORDING-A-POSE.md`

Written: 2026-08-09, after `Parked` was saved with no path and could not be played

① One-page cheat sheet (the values that absolutely must be right)

What	Value	Source
The real emergency stop	the KCD1 rocker switch and the inline fuse. Not the screen.	[<code>arm-console.html</code> header comment]
Hand position on every power change	under the forearm before the rocker — it swings down through the base area where the loom and blue cable sit	[<code>arm-poses.csv</code> , <code>storage</code> notes]
J0 Base travel	<code>0–180°</code> , home <code>90</code> — pin not verified on this arm	[<code>joint-limits.csv</code> J0; <code>calibration-log</code> 2026-08-01]
J1 Shoulder travel	<code>0–91°</code> , home <code>1</code> (not 0, on purpose)	[<code>joint-limits.csv</code> J1, locked 2026-08-05 19:04]
J3 Elbow travel	<code>0–30°</code> , home <code>15</code> — re-locked 2026-08-08, down from 0–66	[<code>joint-limits.csv</code> J3, locked 2026-08-08 17:48]
J4 Wrist pitch travel	<code>33–180°</code> , home <code>90</code> — travel ends still unconfirmed	[<code>joint-limits.csv</code> J4]
J5 Wrist roll travel	<code>0–180°</code> , home <code>90</code>	[<code>joint-limits.csv</code> J5]
J6 Gripper travel	<code>0–180°</code> , home <code>90</code> — does not articulate yet	[<code>arm-poses.csv</code> , <code>pick</code> notes 2026-08-06]
Park margin from any limit	at least 3° inside . A joint on its own end has nowhere to back off to.	[<code>joint-limits.csv</code> — J1 homes at 1 not 0, J3 at 15 not 30]
Firmware speed ceiling	<code>90 deg/s</code> . Above it the firmware answers <code>ERR E12</code> .	[<code>arm-poses.csv</code> , <code>pick</code> notes, 11 cycles 2026-08-06]
Smallest movement that becomes a step	<code>2°</code> . Less is a nudge and is deliberately not recorded.	[<code>arm-console.html</code> <code>AUTO_STEP_MIN_DEG</code>]
Preflight, before servo power	<code>powershell -NoProfile -ExecutionPolicy Bypass -File Documentation\bench-preflight.ps1</code>	[commit 9fb45bc; 8/8 on 2026-08-09]
Rollback the console, no restart	on the Pi: <code>cp ~/arm/console/arm-console.html.bak-* ~/arm/console/arm-console.html</code> , then F5	[<code>arm-bridge.py</code> re-reads the file on every page load]

What	Value	Source
Pull poses back into git afterwards	Software/arm-console/sync-poses.sh	[sync-poses.sh header]

The on-screen E-STOP is not a safety device

Neither is the Arduino, the firmware, nor the browser. A USB cable, a browser tab and a Python program can all die in the middle of a move. The only stop you can rely on is removing servo power at the rocker switch. Nothing on this arm observes the output shaft, so every number on screen is what the console *commanded* — judge the arm with your eyes, not the display.

⚠ 3 things are different from your last bench session — read before you start

1. **You no longer press RECORD WAYPOINT.** Letting go of the jog bar records the step. The old flow made recording a separate act to remember between every jog, and forgetting once wrote a pose that looked saved and could never be played — which is exactly what happened to `Parked` on 2026-08-09. The button still exists; you should not need it. [commit 1565b44]
2. **The console is now four tabs, one joint at a time.** TEACH is where you land. The six joints are a rail down the left; click one and its full card opens in the middle. The envelope sliders, LOCK THIS AXIS, deg/s and LOAD LIMITS FILE moved to CALIBRATION — they are not gone, just out of the way of RECORD. [commits 2630d85, 1b4fc41]
3. **All three saved poses are unusable, and the library now says so.** `storage` (J3=64) and `pick` (J3=36) read **OUT OF RANGE**: the elbow lock narrowed to 0–30 on 2026-08-08, so the arm cannot reach either any more and the firmware would clamp them and stop somewhere that is not the pose. `Parked` reads **REFERENCE ONLY** — no path was recorded. All three stay as history; none is a target. [arm-poses.csv; joint-limits.csv J3]

1 What a pose is — the 60-second mental model

A pose is a named position you can send the arm back to: park, start, pick, handoff. It is **not** six numbers.

Two positions can both be perfectly safe while the straight line between them drives the claw through the bench, the wire loom, or the base housing. So a pose is **the route you drove, in the order you drove it** — the console calls each stop along the way a *step*.

The steps come from the angles the arm was actually commanded to, at each point where you stopped and looked at it. Nothing is guessed and nothing is interpolated. That is why letting go is the moment a step is taken: letting go is when you stopped and looked.

Why this matters: A pose saved with no steps records only where the arm ended up. The library shows it as **REFERENCE ONLY** and refuses to load it, because there is no safe way to drive somewhere when nobody recorded how you got there.

One extra step is taken for you: **the position the arm is standing in when you first touch the jog bar**. Without it, playback would jump straight to step 1 from wherever the arm happens to be — every joint moving at once

— which is the exact hazard above.

2 Phase 0 — everything below runs with servo power OFF

Nothing in this phase moves the arm. The console is built to be exercised end to end with zero servos powered: the board answers, the controls respond, the firmware acknowledges.

- ☐ **P0.1** Run the preflight from the repository root:

```
powershell -NoProfile -ExecutionPolicy Bypass -File Documentation\bench-preflight.ps1
```

✓**You should see:** 8 passed, 0 failed. It confirms the Pi is serving *exactly* the console in the repo, that the pose file matches, and that the hold daemon is stopped.

If it says the hold daemon is **RUNNING**, stop

`hold_arm_pi.py` reports `EN=1` for joints whose supply rail is dead. Switching servo power on with it running snaps every joint to the setpoint it was last holding. Stop it before the rocker goes anywhere near ON.

- ☐ **P0.2** Double-click **Arm Console** on the desktop. A black window opens and stays open — that window is the connection to the Pi. The browser follows a few seconds later.

✓**You should see:** The **TEACH** tab selected, and six joints listed in the left rail.

- ☐ **P0.3** Click through **RUN POSES**, **CALIBRATION**, **DIAGNOSTICS**, and back to **TEACH**.

✓**You should see:** The red **E-STOP** and the safety line stay on screen on all four tabs. They are outside the tabs on purpose.

- ☐ **P0.4** Press **CONNECT**.

✓**You should see:** The status chip turns green and reads *Connected*. No red notices.

- ☐ **P0.5** Go to **CALIBRATION** and click **J3 Elbow**.

✓**You should see:** The envelope reads `0°` and `30°`, and the pill beside ENVELOPE says **ACKNOWLEDGED**.

P0.5 is the gate. Do not pass it.

DEFAULT instead of **ACKNOWLEDGED** means the controller has not confirmed the limits and `joint-limits.csv` never loaded — every joint is on the firmware's `70-110°` placeholder, which is not this arm's travel. Opening the serial port resets the board and the firmware keeps nothing across a reset, so the file is re-read on every connect. If it did not, use **LOAD LIMITS FILE** on the **CALIBRATION** tab and check again.

3 Phase 1 — power on, one joint, prove the recording

Hand under the forearm. Then the rocker.

☐ **T1** Select **J1 Shoulder**, press **ENABLE**, type the angle it is sitting at judged by eye, press **CONFIRM**.

✓**You should see:** Nothing jumps. The first pulse is exactly the value you typed, which is the whole point of being asked.

☐ **T2** The teach table is empty and reads “No steps yet” under SAVE POSE. **Touch the jog bar and hold it still for a second.**

✓**You should see:** A row labelled **start** appears **before the arm has gone anywhere**.

☐ **T3** Jog about 10°, then let go.

✓**You should see:** A **second row** appears, carrying the new angle.

☐ **T4** Nudge it barely — a degree or so — and let go.

✓**You should see:** **No new row.** Below 2° is a nudge, not a step.

☐ **T5** Hold the jog bar so the joint is moving, and **while still holding it, click a different joint tile in the left rail.**

✓**You should see:** The joint **stops immediately**.

☐ **T6** Untick **Save a step each time I let go**. Jog and release: no row. Tick it back on.

What each one is actually testing

Test	Proves	If it fails
T2	The starting position is captured before the first move. Without it, playback jumps to step 1 from wherever the arm stands — every joint at once.	Stop. Recording is not safe to rely on. Do not go to Phase 3.
T4	A nudge does not become a step, so the table stays readable.	Cosmetic. Carry on; delete spare rows with DELETE.
T5	Switching joints releases the one you were holding. Otherwise it keeps taking jog commands at a joint whose controls you can no longer see.	Press E-STOP. Report it. Do not continue.

Delete the practice rows before Phase 2.

4 Phase 2 — make the park pose

Park is where the arm rests with the power off. It is the one pose worth being fussy about.

Rules, not angles

No angles are prescribed here on purpose. The elbow's travel is $0-30^\circ$ now, not $0-66^\circ$ — the fold that storage recorded does not exist any more, and nobody has looked at the arm's geometry since that re-lock. Judge it with your eyes.

- **Stay at least 3° inside every limit.** A joint against its own end has nowhere to back off to. `joint-limits.csv` does this itself: J1 homes at 1° and not the locked 0° , J3 at 15° and not 30° .
- **Fold over the base** rather than reaching out over the bench.
- **Leave the wrist alone** unless you are sure of its travel. J4's ends are still unconfirmed, and driving a wrist blind is how the claw gets pushed into the mat.
- **Think about power-off.** The forearm swings down through the base area where the loom and the blue cable sit.

☐ **T7** Drive there in $8-12^\circ$ steps, letting go at each one. Watch the arm, not the screen.

☐ **T8** Read the line above **SAVE POSE**.

✓You should see: "Will save N steps as the path to this pose."

If it still says "No steps yet", do not save

Nothing was recorded, and saving would produce another `Parked`: a row that looks saved and can never be played. Go back and jog with the tick box on.

☐ **T9** Type `park` in the name box and press **SAVE POSE**.

✓You should see: A green notice naming the pose and its step count.

☐ **T10** Check for an amber warning about a joint on a soft limit.

✓You should see: **No amber warning.** If you get one you are parked on an end — back off a few degrees and save again. It warns and still saves; parking on a limit deliberately is your call, just not one to make by accident.

☐ **T11** Go to **RUN POSES**.

✓You should see: `park` shows a green **READY** badge with its step count, and its **LOAD** button is enabled.

5 Phase 3 — the first playback

Nothing taught on this arm has ever been played back. Go slowly, and keep a hand free for the rocker.

☐ **T12** Jog the arm away from park, somewhere obviously different.

☐ **T13** On **RUN POSES**, press **LOAD** on park .

✓**You should see:** You land on **TEACH** with the recorded steps in the table. Read them before going further — that review is the reason **LOAD** and **PLAY** are two buttons.

☐ **T14** Press **PLAY**.

✓**You should see:** The arm retraces the path step by step and ends at park.

Abort — E-STOP or the rocker — if any of these happen

- any joint reports **CL=1** (clamped) or **JT0=1** (joint timeout)
- the watchdog latches, or a joint detaches on its own
- anything moves that you did not expect, or in an order you did not expect
- the claw heads toward the bench, the mat, the loom or the base housing

If it aborts, write down which step number it was on. The step number plus the table is the whole diagnosis.

If T14 is clean, the run has succeeded

Phase 4 is optional. Stop here with a working park pose and you have got what this session was for.

6 Phase 4 — optional: re-record the two broken poses

`storage` and `pick` are marked **OUT OF RANGE** because the elbow lock narrowed from `0-66°` to `0-30°` after they were recorded. The arm cannot reach either any more.

☐ **T15** Teach and save `storage-v2`, the same way as `park`.

☐ **T16** Teach and save `pick-v2`.

Why this matters: The old rows stay exactly where they are. `arm-poses.csv` is append-only history: a row was true on the date it was written, and deleting it would erase the record of what the arm used to do. They are simply not targets any more.

7 Phase 5 — get the poses into git

Do this before closing anything. The pose is written into `arm-poses.csv` on the Pi, next to the bridge — which is a deploy copy, not a git checkout. A pose that lives only there is invisible to git and to anyone else working on this arm.

```
Software/arm-console/sync-poses.sh
git diff Software/arm-console/arm-poses.csv
```

☐ **T17** Read the diff.

✓**You should see: Additions only.** An edited or removed row means something went wrong — the file is append-only.

☐ **T18** Commit it.

8 Powering down

☐ Hand under the forearm **before** the rocker. It swings down through the base area on power-off.

☐ Rocker off.

☐ Leave the hold daemon stopped. Do not start it “to hold position” — it reports `EN=1` for joints whose rail is dead, so it is exactly what makes the next power-on dangerous.

9 Record the run

Fill this in and hand it back. Blank is a result — “did not get to it” is worth knowing.

Date / time: _____

Preflight: pass / fail which check _____

Phase 1 T1-T6: pass / fail which test _____
what happened: _____

park pose saved: yes / no steps recorded: _____

Playback T14: clean / aborted at step ____ / not tried

storage-v2: yes / no

pick-v2: yes / no

Synced and committed: yes / no

Anything the console SAID that turned out not to be true:

That last line is the most valuable one on the page

The console now makes claims before you act: how many steps it will save, whether a pose is READY, whether a joint is on its limit. Those claims have been tested with no board attached, but a bench is not a test harness. If any of them turned out to be false in front of the real arm, that is worth more than the pose.